

This Paper

Aim to measure gender bias among colleagues in an anonymous online setting where members of the economics profession are presumably freed from social pressure.

Two main goals:

- ▶ Quantify gender stereotyping in a non-lab setting
- ▶ Demonstrate the use of text analytic methods in economics research

Background on EJMR

- ▶ Established to share job market information: Professional & Academic purposes
 - ▶ 306,253 number of threads available on the EJMR as of Oct 2017, in a span of seven years; 391,720 as of Nov 2019
- ▶ **Anonymity** presumably removes the pressure to edit one's speech. More likely to capture what one believes/thinks but may not openly say.
- ▶ Moderation policy - automatic censorship & reports by users

Economics Job Market Rumors					Chat (0)	Conferences	Forums	Job Market	Journals
					8552.07 BTC/USD (0.18 UKAP)				
Topic — Filter — New Topic »					Posts	Views	Votes	Freshness	
Against Economics - 2 3					57	1618	5-2	11 hours	
Imagine if one day Kirk receives a subpoena in a class action					1	79	0-0	11 hours	
Best introductory sociology textbooks					3	121	0-0	11 hours	
Fabulous Fab: HRM job at great location; U: hoping for a LAC bumblefcuk flyout					10	580	0-0	11 hours	
I'm co-authoring a paper with my ex-bf					3	219	0-0	12 hours	
Sociology using National Experiments					5	192	0-0	12 hours	
EJMR is basically the perfect forum for an aspie fvktard					5	155	1-2	12 hours	

Outline

- ▶ EJMR Data
- ▶ Measure Gendered Language - suggestive of an unwelcoming/toxic online environment for women
- ▶ Topic Analysis - (1) Professional, (2) Personal
 - ▶ Cross sectional: are *Female* and *Male* posts different on average?
 - ▶ Panel / Dynamic approach: how does gender affect the **transitions** between topics within a thread?
- ▶ A Model of Rumors: why do people post differently about women versus men?
- ▶ Tips on text analysis

Data

Scraping:

- (1) Information about the **Threads**: title, a rough timestamp of the most recent post, number of posts, number of views, votes
 - ▶ > 500K threads as of Nov 2021
 - ▶ 223K threads initiated or updated from 2013 to 2017
- (2) **Posts** on the first and the last page (if > 1) of each thread in the past 4 years.
 - ▶ 2,217,046 posts across 223,475 threads

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Data

- ▶ Most frequent 10,000 words in EJMR posts: use token counts as input variables in text analysis
- ▶ Identify **gendered** posts that discuss about women or men
 - ▶ classifiers from the most frequent 10,000 words: “he”/“she”, “man”/“woman”, “girl”/“boy”
 - ▶ names: gendered first names, or names of economists
 - ▶ what if a post includes both female & male classifiers? → Lasso exercise
- ▶ Identify **job ranks** at the post level: graduate students, job market candidates/post-docs, junior faculty (untenured), senior faculty (tenured)

Data

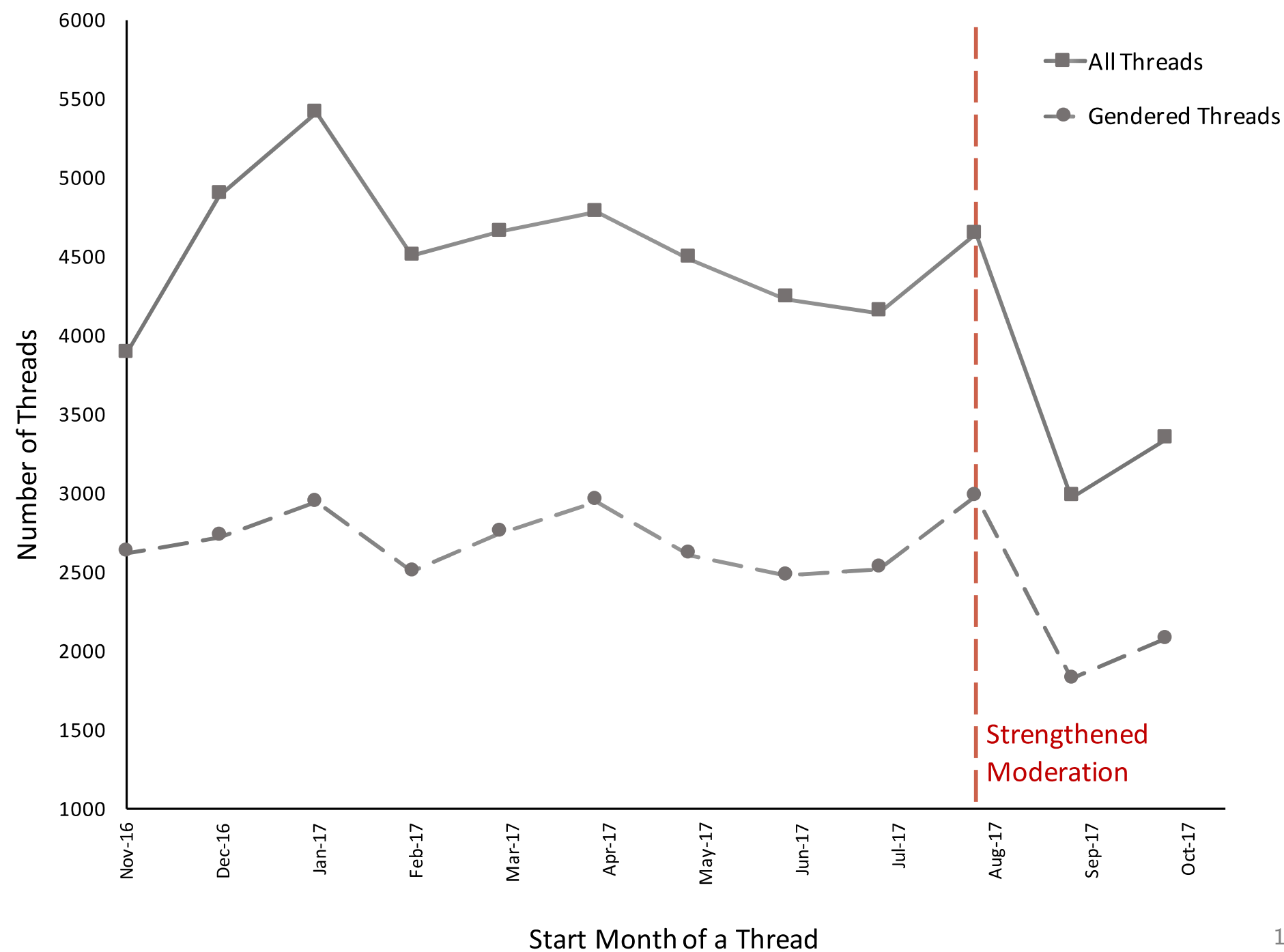


Table: Sample Overview

	Female	Male
<i>Before August 2017</i>		
No. Posts	99,659	318,873
No. Threads	41,243	116,996
<u>No. Posts by Job Rank</u>		
Graduate Students	3,111	11,359
Job Market Candidates/Post-docs	2,156	8,932
Junior Faculty	2,335	9,675
Senior Faculty	2,097	18,339
<i>August - October 2017</i>		
No. Posts	4,817	15,848
No. Threads	2,129	6,192
<u>No. Posts by Job Rank</u>		
Graduate Students	181	639
Job Market Candidates/Post-docs	112	374
Junior Faculty	159	718
Senior Faculty	131	1,059

Gendered Language

Two goals of a predictive model:

- ▶ Resolve posts that cannot be classified by gender classifiers alone
- ▶ Identify words that are most predictive about each gender, which reveals a relationship between the gender of the subject and the characteristics posters wish to highlight

Lasso-Logistic Model

For each post i ,

- ▶ $Female_i = 1$ if it discusses a female, and 0 if it discusses a male.
- ▶ $\mathbf{w}_i$: a vector of counts for each of the most frequent 10,000 words except for the gender classifiers that are present in the post i .

Logistic regression:

$$P(Female_i = 1 | \mathbf{w}_i) = \frac{\exp(\theta_0 + \mathbf{w}_i' \boldsymbol{\theta})}{1 + \exp(\theta_0 + \mathbf{w}_i' \boldsymbol{\theta})}$$

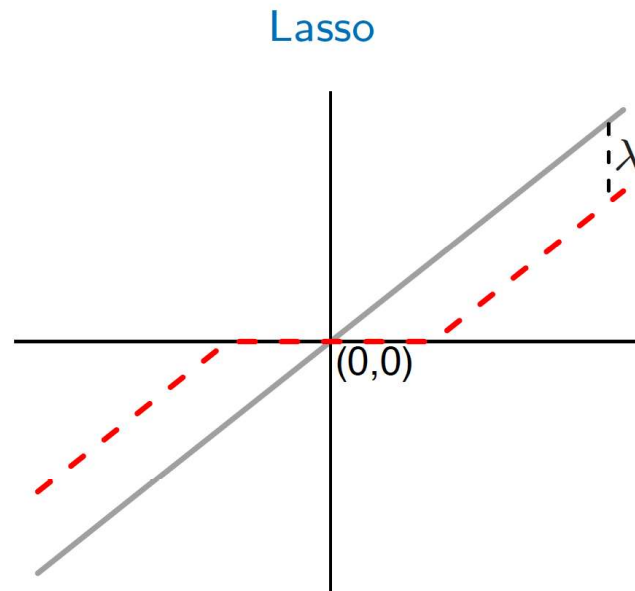
$$P(Female_i = 0 | \mathbf{w}_i) = 1 - P(Female_i = 1 | \mathbf{w}_i)$$

Lasso-Logistic Model

$$P(\text{Female}_i = 1 | \mathbf{w}_i) = \frac{\exp(\theta_0 + \mathbf{w}_i' \theta)}{1 + \exp(\theta_0 + \mathbf{w}_i' \theta)}$$

Lasso-regularized model → identifying words that predict F/M posts while avoiding overfitting.

$$\hat{\theta}_\lambda = \operatorname{argmin}_\theta -\log(\prod_{i=1}^N P(\text{Female}_i | \mathbf{w}_i)) + \lambda \|\theta\|_1$$



Lasso-Logistic Model

$$P(\text{Female}_i = 1 | \mathbf{w}_i) = \frac{\exp(\theta_0 + \mathbf{w}_i' \boldsymbol{\theta})}{1 + \exp(\theta_0 + \mathbf{w}_i' \boldsymbol{\theta})}$$

Object of interest: average marginal effect of each word on Female_i

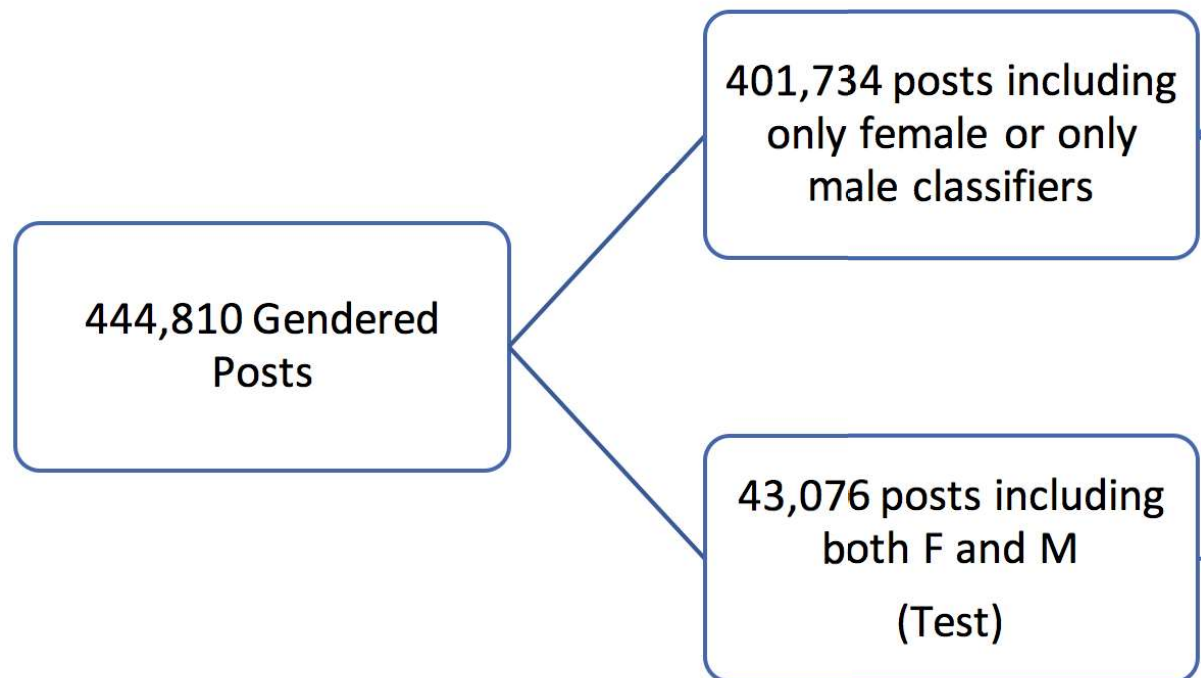
Letting w_i^k denote the number of word k in post i , we have

$$\frac{\partial}{\partial w_i^k} \Pr(\text{Female}_i = 1 | \mathbf{w}_i) = \theta_k * \Pr(\text{Female}_i = 1 | \mathbf{w}_i) * \Pr(\text{Female}_i = 0 | \mathbf{w}_i)$$

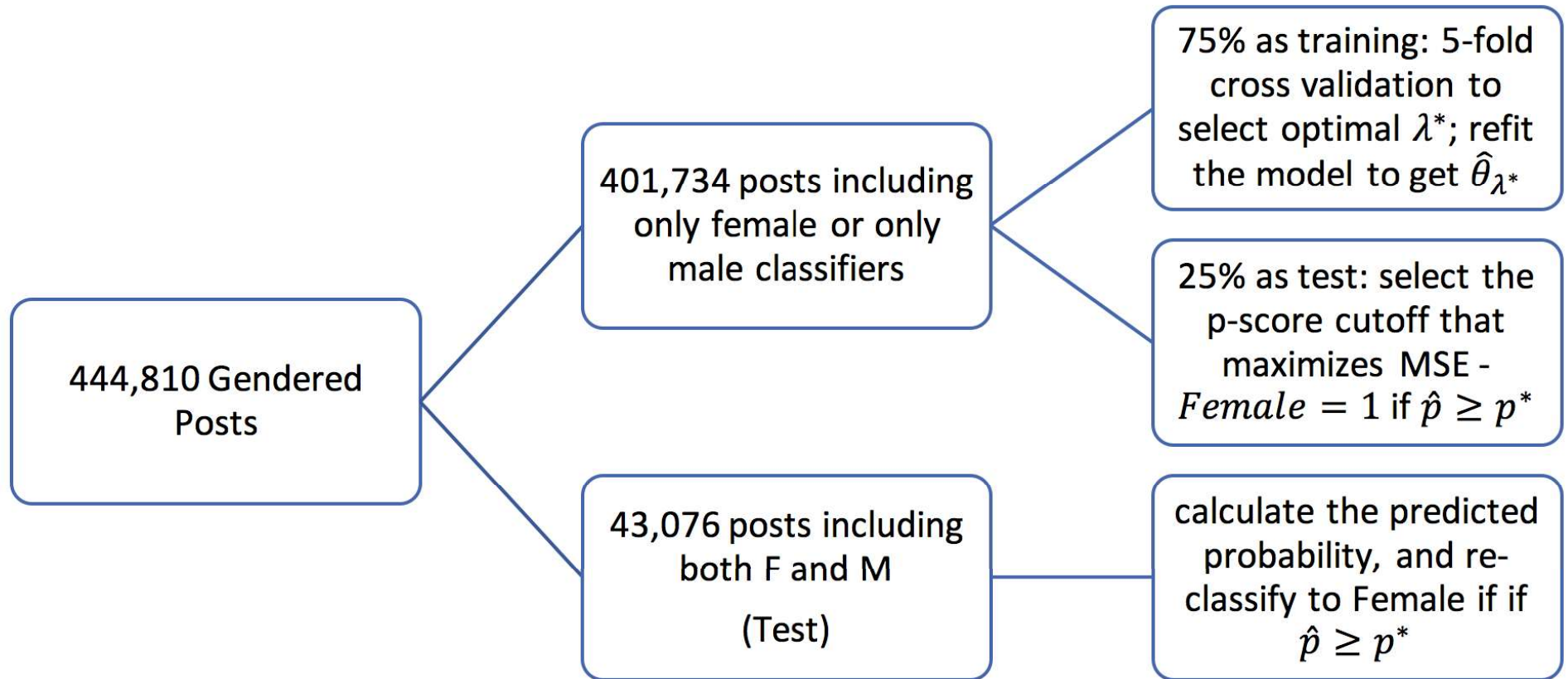
Therefore, the **average marginal effect** of word k across N posts is:

$$\frac{1}{N} \sum_i \Pr(\text{Female}_i = 1 | \mathbf{w}_i) * \Pr(\text{Female}_i = 0 | \mathbf{w}_i) * \theta_k$$

Model Training



Model Training

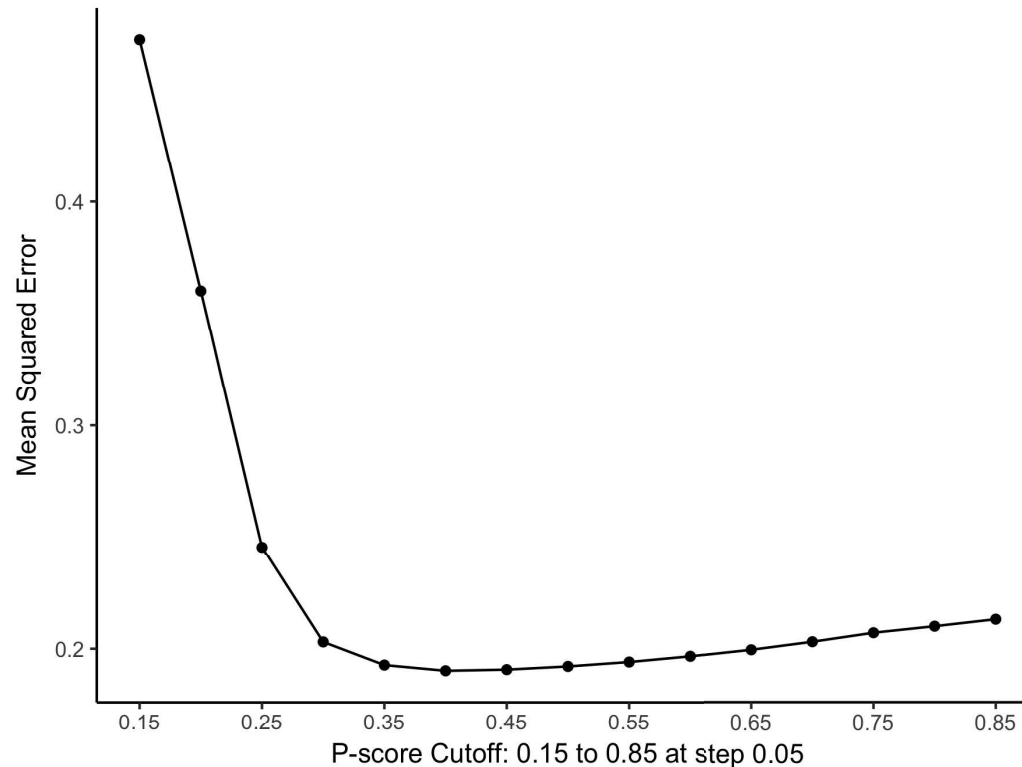


Note:

- ▶ Tuning parameter λ^* maximizes the average accuracy score across 5 folds
- ▶ P-score cutoff p^* from reserved test set: see figure below

Model Training

Figure: Selection of Optimal P-score Cutoff by Mean Squared Error



Note: “P-score” refers to the predicted probability of the subject of a post being female, i.e. $P(\text{Female} = 1 | \text{Words})$. The MSEs are calculated on the test set - 25% of all posts that include only “female” or only “male” words. $p = 0.40$ is selected as the optimal cutoff to make a call on gender among the ultimate test set.

Gendered Language

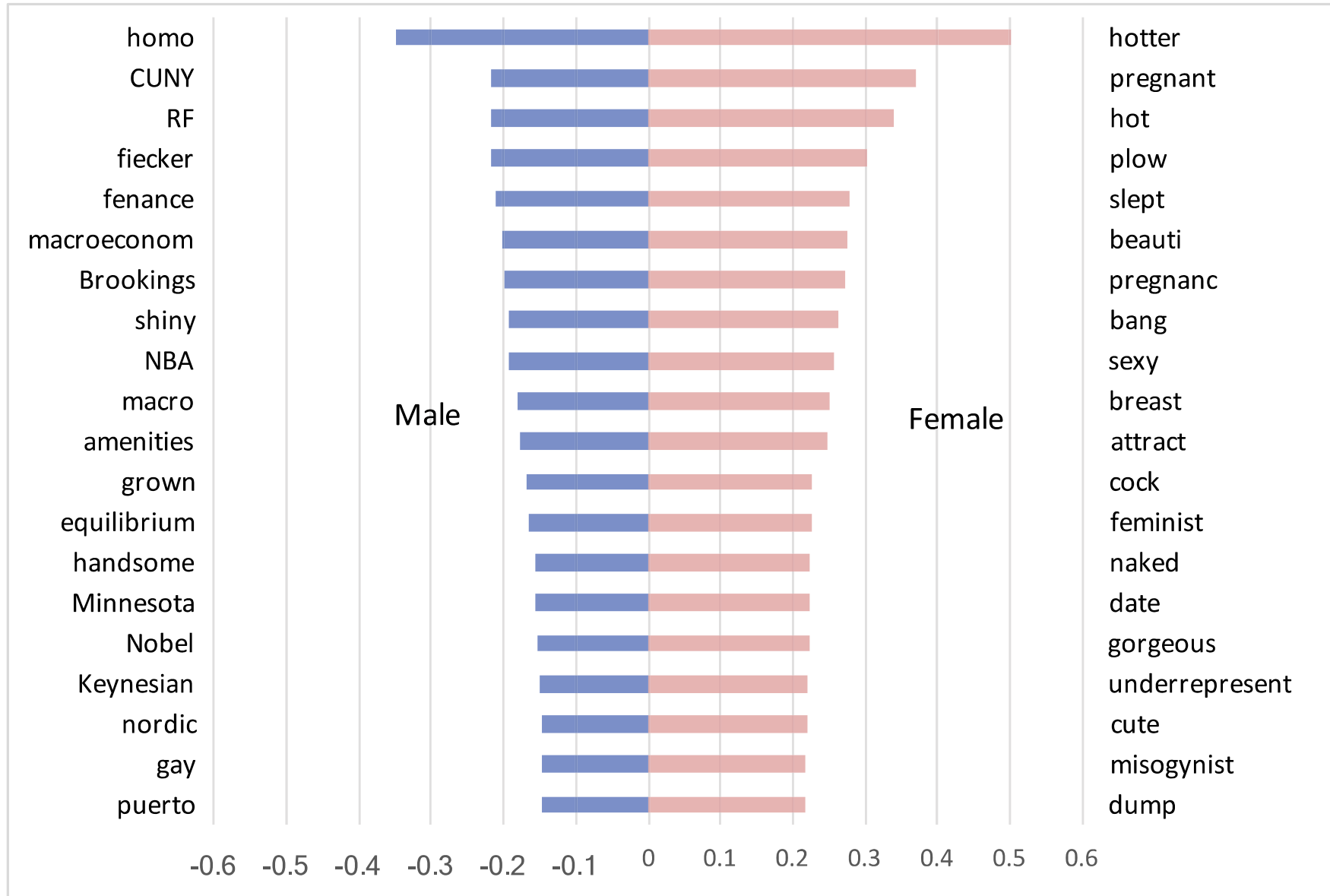
Table: Top 10 Words Most Predictive of Female / Male

Most <i>female</i>		Most <i>male</i>	
Word	Marginal Effect	Word	Marginal Effect
hotter	0.422	homo	-0.303
pregnant	0.323	testosterone	-0.195
plow	0.277	chapters	-0.189
marry	0.275	satisfaction	-0.187
hot	0.271	fieckers	-0.181
marrying	0.260	macroeconomics	-0.180
pregnancy	0.254	cuny	-0.180
attractive	0.245	thrust	-0.169
beautiful	0.240	nk	-0.165
breast	0.227	macro	-0.163

Notes: The marginal effect of word w is the change in probability that a post is discussing a female, when it contains an additional w . The model is trained on non-duplicate gendered posts identified by any classifier. (frequency)

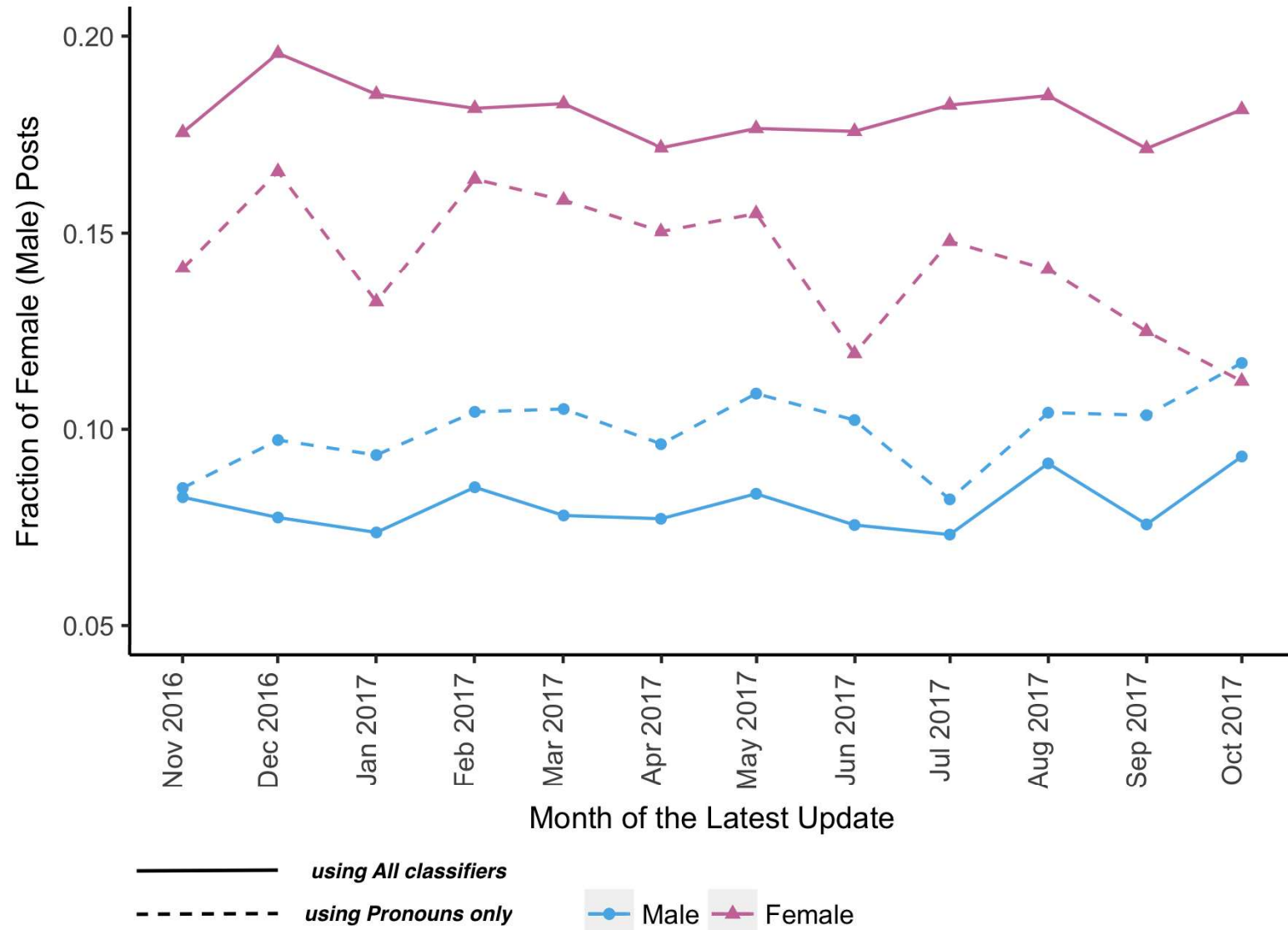
Gendered Language

Top 20 words after word stemming (nltk.stem):



Are the Most Predictive Words Frequent?

Figure: Fraction of *Female* (*Male*) Posts that include Any Top 50 *female* (*male*) Words, by Month of the Latest Post



Double Check: Most Frequent Words

Table: Most Frequent Words in Female / Male Posts

Most common in <i>Female</i>			Most common in <i>Male</i>		
Word	No. <i>Female</i>	No. <i>Male</i>	Word	No. <i>Female</i>	No. <i>Male</i>
life	4,034	7,644	work	3,800	13,989
work	3,800	13,989	paper	1,503	11,727
hot	3,613	1,053	job	3,091	10,313
love	3,297	4,274	economics	1,120	9,808
sex	3,103	1,535	great	2,323	9,181
job	3,091	10,313	best	2,558	8,552
feel	2,574	5,167	research	1,407	8,238
best	2,558	8,552	school	2,446	8,228
school	2,446	8,228	market	1,750	7,954
kids	2,441	2,200	life	4,034	7,644

Notes: The words that are most common in *Female* (*Male*) are sorted by the number of *Female* (*Male*) posts they appear in.